Aakarshak Swaraj

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PROFESSIONAL SUMMARY

AI/ML Engineer with 2 internships and 7+ production-grade projects in deep learning, NLP, and computer vision. Improved medical image classification to **93.4% accuracy** using ResNet50 with SimCLR contrastive learning, elevated sports prediction to **AUC 0.81** via ensemble methods, and drove **95% precision** on a transformer-based NLP pipeline deployed for civic monitoring. Skilled in Python, PyTorch, Hugging Face Transformers, OpenCV, scikit-learn, model optimization, and prompt engineering.

EDUCATION

Vellore Institute of Technology August 2023 – August 2027 (Expected)
B.Tech in Computer Science, Specialization in AIML; GPA: 8.62/10.0 *Chennai, India*

WORK EXPERIENCE

SDE Intern | Taslasoft | Hyderabad, India February 2026 – April 2026
- Developed a real-time social intelligence pipeline in **Python** monitoring **15+ subreddits** and analyzing **10,000+ posts daily**, mapping incident activity to geographic clusters through a geo-temporal aggregation system with sub-second latency.
- Reached **95% precision** on a 3-stage NLP detection pipeline (keyword filtering → sentiment analysis → Hugging Face transformer classification) with human-in-the-loop validation, flagging **200+ harmful posts** during the pilot deployment.

SDE Intern | Qcecuring Technologies | Delhi, India October 2025 – December 2025
- Architected a self-service code-signing platform with client-side hashing, reducing key-exposure incidents to **zero** and cutting artifact signing time by **60%** compared to the previous manual workflow.
- Delivered a **PKI-as-a-Service** module supporting **50+ enterprise certificates**, with quorum-based multi-approver workflows, automated lifecycle management, and tamper-proof audit logging for compliance.

PROJECTS

Breast Cancer Detection | *Python, PyTorch, ResNet50, SimCLR, Deep Learning*
- Fine-tuned **ResNet50** with SimCLR contrastive pretraining on **7,909 BreakHis** histopathology images, reaching **93.4% accuracy** and **0.92 F1-score** — a ∼7% gain over the CNN baseline — using PyTorch with augmentation-heavy preprocessing.

FIFA World Cup 2026 Prediction GitHub | *Python, Scikit-learn, NumPy, SciPy, Matplotlib*
- Built a multinomial logistic regression model fusing forward-in-time **Elo ratings** with squad-strength features across **900+ international matches**; ran **20,000 Monte Carlo** simulations to forecast title probabilities, backtested on 2018 and 2022 World Cups with temporally-correct splits and **log-loss below baseline**.

TSlate – Offline Voice Translator GitHub | *C++, Kotlin, Whisper.cpp, NLLB-200, CTranslate2, JNI/NDK*
- Engineered a fully offline Android voice translator (Hindi/Hinglish → Kannada) targeting **<15s** end-to-end latency by integrating Whisper.cpp (ggml-tiny) for STT and **NLLB-200 600M INT8** for neural translation via custom C++ JNI/NDK bindings with Unicode-based Devanagari/Latin script routing.

IPL Match Prediction & Decision Support GitHub | *Python, Scikit-learn, SQLite, Pandas, NumPy*
- Designed an IPL outcome predictor scoring **AUC 0.81** across 1,169 matches (2008–2026) using a tuned soft-voting ensemble with 20+ engineered features and an SQLite-backed incremental retraining pipeline; implemented a constrained **Playing XI optimizer** enforcing role composition and overseas limits.

VARUNA – Disaster Reporting Platform (SIH 2025) GitHub | *Next.js, Python, OpenCV, NLP*
- Led a **5-person team** to build VARUNA, processing **5,000+ monthly reports** with a multi-stage CV+NLP validation pipeline that reduced false submissions by **∼35%**; deployed OpenCV for image verification and transformer-based text classification for report triage.

EYERIS – Privacy-Preserving Autonomous Robot GitHub | *Python, OpenCV, MediaPipe, Arduino*
- Deployed a privacy-preserving robot with **<50ms** real-time human detection latency via MediaPipe pose estimation and OpenCV, eliminating camera-based identification while maintaining a **∼30cm** obstacle avoidance threshold across **3 lab environments**.

TECHNICAL SKILLS

ML & AI: PyTorch, Scikit-learn, Hugging Face Transformers, OpenCV, NLP, Computer Vision, LLMs, RAG, Fine-Tuning, Prompt Engineering, Model Optimization (Quantization, ONNX)
Languages: Python, C++, JavaScript, Kotlin, Dart **Data:** Pandas, NumPy, SciPy, Matplotlib, MLflow
Backend & Systems: FastAPI, Node.js, Spring Boot, REST APIs, Distributed Systems
Databases: PostgreSQL, MySQL, MongoDB, SQLite **Cloud & DevOps:** AWS (EC2, S3, Lambda), Docker, CI/CD
Tools: Git, Linux, Maven, Gradle, CTranslate2, Jupyter

CERTIFICATIONS

- **Anthropic AI Courses** Certificate — May 2026
- **Generative AI Using IBM Watsonx** – IBM Certificate — Jun 2025
- **Supervised Machine Learning: Regression and Classification** – DeepLearning.AI Certificate — Dec 2024

ACHIEVEMENTS

- **Gold Level – WorldQuant BRAIN International Quant Championship 2026:** Ranked among **top 20% of teams globally** in a quantitative research challenge. Jun 2026
- **SIH 2025 – VARUNA:** Shortlisted from **864 teams** to top 150 nationally; ranked **51st** in second round. Sep 2025
- **National Semi-Finalist, Flipkart GRID 7.0** – Techno-commercial hackathon by Flipkart Early Careers. Aug 2025
- **National Finalist, 3rd Runner-up – DATASET '24:** Shipped an AI system in 36 hours; earned a **Nokia internship offer**. Dec 2024